

RBI Monetary Policy Communication Analysis

A 10-Year Corpus Study (2016–2026) Using NLP and Event Study Methodology

Methodology and Model Description – Project Report

1. Project Overview

This project investigates how the Reserve Bank of India (RBI) communicates monetary policy and whether the sentiment embedded in its official communications systematically influences equity market returns. Using a decade-long corpus of Monetary Policy Statements (MPS) and Post Monetary Policy Press Conferences (PMPPC) spanning April 2016 to June 2026, the study applies a three-stage NLP pipeline – topic modelling, sentiment classification, and event-study regression – to quantify the market impact of central bank communication tone.

The core research question is: does the hawkish or dovish tone of RBI communications, decomposed by policy topic, generate statistically significant abnormal returns in the BSE Sensex, and over what time horizon does this effect persist?

2. Data Collection and Corpus Construction

2.1 Source and Scope

Official documents were scraped directly from the RBI website (rbi.org.in) using Python's requests library and BeautifulSoup for HTML parsing. Two document types were collected:

- Monetary Policy Statements (MPS): formal post-meeting communiqués issued by the Monetary Policy Committee (MPC) articulating rate decisions, stance, and forward guidance.
- Post Monetary Policy Press Conferences (PMPPC): transcripts of Q&A sessions between the Governor/Deputy Governors and the media, capturing real-time elaborations.

The final corpus spans 51 MPC meeting cycles (April 2016 to June 2026), yielding 102 documents and 2,336 paragraph-level text units after sentence tokenisation and cleaning. For press conferences, speaker attribution logic filtered responses to MPC officials only (Governor, Deputy Governors), excluding journalist questions.

2.2 Text Preprocessing

Each raw HTML document underwent the following preprocessing steps: **(i)** footnote/citation removal via regex, **(ii)** whitespace normalisation, **(iii)** NLTK sentence tokenisation, and **(iv)** minimum-word filtering (sentences with fewer than 3 words were discarded) to remove artefacts such as standalone dates or headings.

The cleaned corpus contained 2,336 rows with metadata fields: date, source_type, speaker, paragraph_id, and text. This paragraph-level granularity is essential for topic-conditioned sentiment aggregation in later stages.

3. Methodology

3.1 Stage 1 – Topic Modelling with BERTopic

To decompose communication into policy-relevant thematic streams, BERTopic was applied – a modular topic modelling framework that combines transformer-based sentence embeddings with clustering algorithms.

Embedding Layer

Each paragraph was encoded into a 384-dimensional dense vector using the all-MiniLM-L6-v2 Sentence Transformer (a distilled BERT variant optimised for semantic similarity tasks). This model was chosen for its strong performance on sentence-level financial text and its computational efficiency relative to full BERT.

Dimensionality Reduction

UMAP (Uniform Manifold Approximation and Projection) reduced the 384-dimensional embeddings to 5 dimensions with parameters `n_neighbors=15`, `min_dist=0.0`, and cosine distance metric. UMAP was selected over PCA because it preserves local topological structure, which is critical for identifying semantically coherent clusters.

Clustering

HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) performed cluster assignment on the UMAP-reduced space with `min_cluster_size=150` and `min_samples=10`. HDBSCAN does not require a pre-specified number of clusters and natively identifies low-density noise points (assigned `topic_id = -1`), making it well-suited to a specialised policy corpus where off-topic paragraphs exist.

Topic Representation

Topic keywords were extracted using a c-TF-IDF weighting scheme. Two representation enrichment models were applied: Maximal Marginal Relevance (MMR, `diversity=0.3`) to reduce keyword redundancy, and a Part-of-Speech filter (nouns and adjectives only) to improve interpretability. BERTopic was configured with `nr_topics=10` as the target, reducing to three interpretable substantive topics after noise removal:

Topic Name	Paragraph Count	Representative Keywords
Liquidity Management and Reserve Operations	515	liquidity, banks, rate, market, billion
Inflation Dynamics and Price Stability	462	inflation, prices, mpc, food, cent
Economic Growth and Demand Dynamics	348	growth, demand, activity, global, production
Other / Press Conference (Noise)	304	question, press, governor, request

Table 1: BERTopic output – topic breakdown across the 2016–2026 corpus.

3.2 Stage 2 – Sentiment Classification with FinBERT

Monetary policy sentiment classification was performed using `yyanghkust/finbert-tone`, a BERT-based model fine-tuned on financial news and analyst reports. For each paragraph, FinBERT returns one of three labels – Positive, Negative, or Neutral – with a calibrated confidence score.

Why FinBERT?

General-purpose sentiment models (e.g., VADER, TextBlob) are calibrated on consumer-facing text and

misinterpret domain-specific financial vocabulary. For instance, the word "tight" is negative in general language but hawkish (signalling restrictive policy) in central bank communication. FinBERT, trained on financial corpora, captures these domain semantics. Compared to alternatives such as Loughran-McDonald dictionaries (lexicon-based, no context), FinBERT's contextual BERT architecture handles sentence-level nuance including negations and hedging language common in RBI statements.

Monetary Policy Label Mapping

FinBERT's financial sentiment labels were mapped to central bank communication taxonomy: Positive → Dovish (accommodative stance), Negative → Hawkish (restrictive stance), Neutral → Neutral. A signed score was computed: +score for Dovish sentences, -score for Hawkish, and 0 for Neutral, enabling a continuous net-sentiment metric.

Running the pipeline over 1,629 substantive sentences (after noise removal) produced the following distribution:

Sentiment	Count	Share (%)
Neutral	914	56.1%
Dovish (Positive)	453	27.8%
Hawkish (Negative)	262	16.1%

Table 2: Overall sentiment distribution across the corpus.

The dominance of Neutral sentences reflects the formal register of central bank language. The Dovish surplus over Hawkish (27.8% vs. 16.1%) is consistent with the broadly accommodative stance of the RBI during much of the sample period, including the post-COVID easing cycle.

3.3 Stage 3 – Sentiment Aggregation and Regression

Aggregation

Sentence-level sentiment was aggregated to the meeting-date × topic level. For each MPC meeting date and topic, the Sentiment Balance Index (SBI) was computed following Gorodnichenko et al. (2023):

$SBI = (Dovish - Hawkish) / (Dovish + Hawkish)$

This yields a score in [-1, +1] per meeting-topic cell, where -1 is fully Hawkish and +1 is fully Dovish. The final regression dataset contains 233 rows (approximately 60 MPC meetings × 4 topics) and serves as the input to the market impact analysis.

Market Impact: Local Projections

BSE Sensex daily closing prices were obtained via the yfinance library (ticker: ^BSESN). For each MPC meeting date t and forward horizon h (h = 0, 1, ..., 30 trading days), cumulative log returns $R_{\{t,h\}}$ were computed as the benchmark for equity market response.

A Local Projections (LP) model was estimated for each topic separately to trace the Impulse Response Function (IRF) of Sensex returns to a unit Dovish shock in that topic's sentiment balance. The specification follows:

$$R_{\{t,h\}} = \alpha + \beta^h SBI_{\{topic,t\}} + \gamma_1 COVID_t + \gamma_2 Rajan_t + \gamma_3 Patel_t + \varepsilon_{\{t,h\}}$$

Control dummies capture structural breaks: COVID (March 2020 – December 2021), Rajan tenure (pre-September 2016), and Patel tenure (September 2016 – December 2018). The model is estimated via OLS (statsmodels) at each horizon h, with bootstrap confidence intervals (500 resamples) to account for the small MPC meeting sample (~60 observations).

4. Key Results

The IRF analysis reveals topic-heterogeneous market responses to RBI communication:

- Inflation Dynamics and Price Stability:** Dovish sentiment in inflation-related communication generates a persistent positive equity return, rising from +16 bps at $h=0$ to +113 bps by $h=23$. This is consistent with markets interpreting reduced inflation concern as a signal of continued accommodation.
- Liquidity Management and Reserve Operations:** Dovish liquidity signals produce a negative market response (−89 bps by $h=8$), potentially reflecting investor concern about financial stability implications of excess liquidity.
- Economic Growth and Demand Dynamics:** Dovish growth communication elicits an initial positive response (+47 bps at $h=0$) that reverses and declines at longer horizons, suggesting short-lived sentiment effects.

These results demonstrate that disaggregating communication by topic substantially improves the signal quality of central bank sentiment measures for market impact estimation.

5. Model Summary and Justification

Model / Tool	Stage	Why Used
all-MiniLM-L6-v2	Topic Embedding	Efficient BERT distillation for sentence-level semantics; strong benchmark performance on financial text
UMAP	Dimensionality Reduction	Preserves local manifold structure; superior to PCA for clustering-oriented pipelines
HDBSCAN	Clustering	Density-based; no fixed k required; native noise identification
BERTopic	Topic Modelling	Modular, interpretable, c-TF-IDF keywords; avoids LDA bag-of-words limitation
FinBERT (finbert-tone)	Sentiment	Domain-specific financial BERT; handles policy language better than lexicon-based or general-purpose models
Local Projections (LP)	Causal Inference	Non-parametric IRF estimation; robust to model misspecification; standard in monetary policy event studies

Table 3: Model architecture and rationale.

6. Limitations and Future Work

The current implementation has several limitations. First, the corpus is limited to English-language documents, excluding Hindi-medium communication that may carry distinct signalling content. Second, the BERTopic run produced only three substantive topics due to HDBSCAN's density requirements; a larger corpus may yield finer-grained topic resolution. Third, FinBERT was fine-tuned on financial news rather than central bank communication specifically; a domain-adapted model (e.g., further fine-tuning on FOMC transcripts) could improve classification precision.

Future extensions include: (i) testing robustness with the FOMC-RoBERTa model as an alternative sentiment classifier; (ii) expanding the dependent variable to Nifty 50 sectoral indices to test heterogeneous market responses; and (iii) constructing a real-time sentiment monitoring dashboard for live MPC communication.